

Team Name : NAVAMESH

Team Leader Name : Rohit Kumar Dash

Problem Statement : AI-Powered Digital Twin of India's Climate using India's National Data

Team Members

Team Leader:

Name: Rohit Kumar Dash
College: Silicon University

Team Member-1:

Name: Subhransu Satapathy
College: Silicon University

Team Member-2:

Name: Shreyash Prusty
College: Silicon University

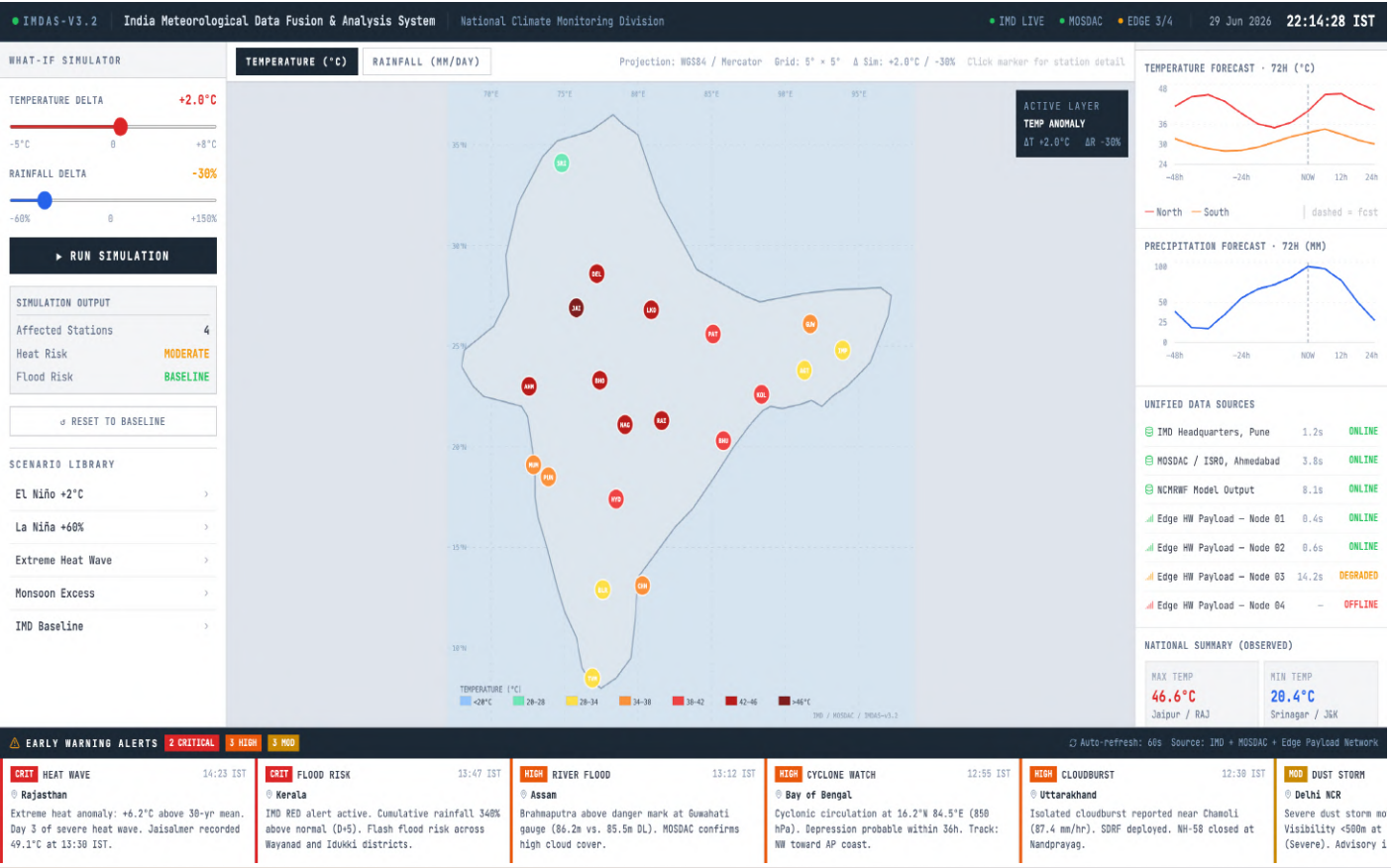
Team Member-3:

Name: Asutosh Kar
College: Silicon University

Opportunity & USP

- **How is it different:** Most entries forecast a single variable from one dataset. Our System fuses ISRO/MOSDAC satellite observations (INSAT LST, SST, rainfall) with IMD ground-based data into one continuously-updating digital twin, paired with an Ai satellite payload concept that feeds back hyper-local, real-time telemetry in the calibrated dashboard.
- **How it solves the problem:** For the PoC, our system ingests and aligns rainfall and temperature data for a selected pilot region, runs it through trained Spatio-temporal DL models (ConvLSTM / LSTM) to generate short-term predictions, and exposes everything through an interactive map dashboard with a "what-if" simulator so officials can test scenarios (e.g., +2°C, -20% rainfall) before they occur, instead of reacting after the fact.
- **USP:** A hybrid hardware + software stack — a scalable national data-fusion and AI pipeline combined with a custom-built edge-computing CubeSat payload concept that supplies independent, real-time ground-truth telemetry to validate and sharpen the twin, going beyond a pure-software dashboard submission.

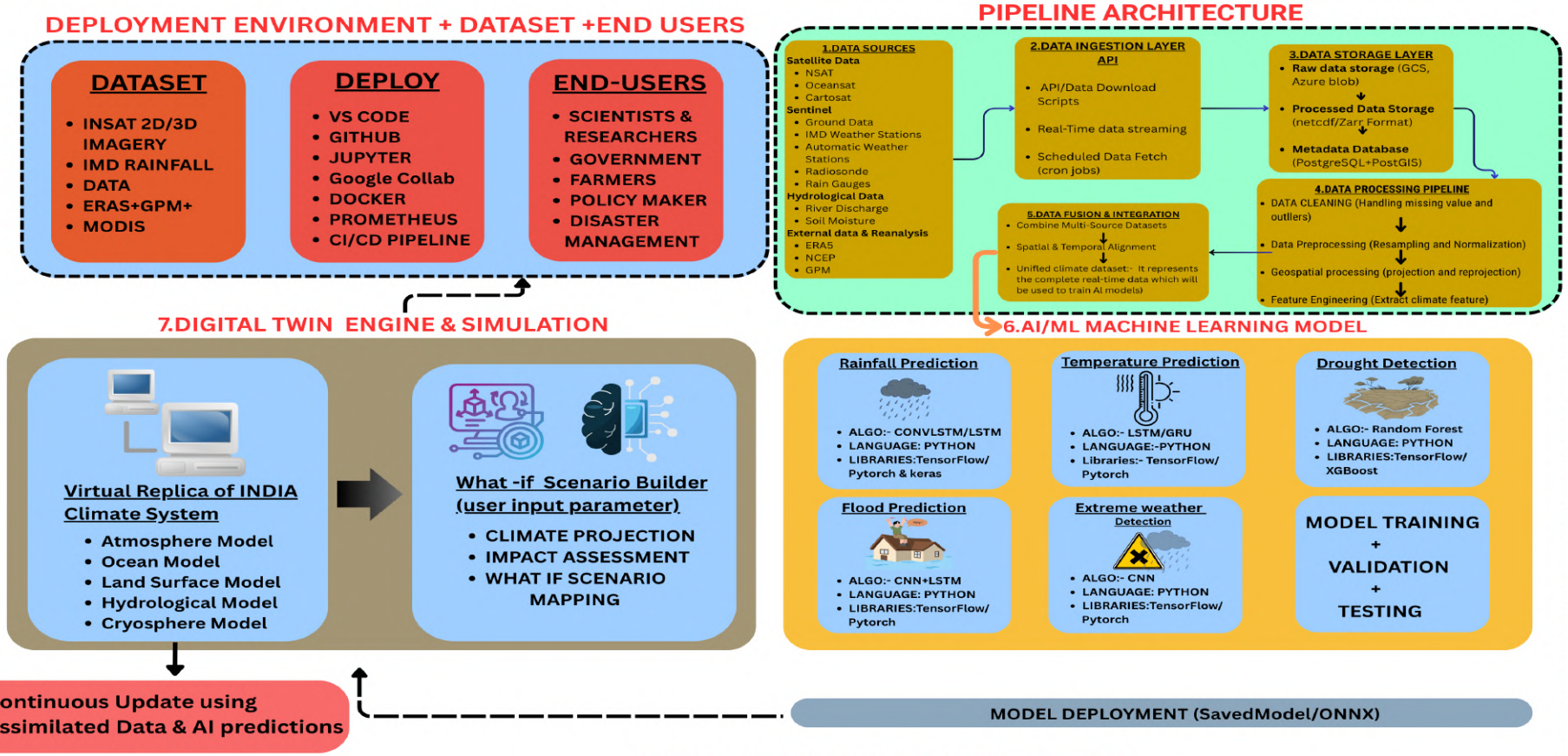
Dashboard



Key Features

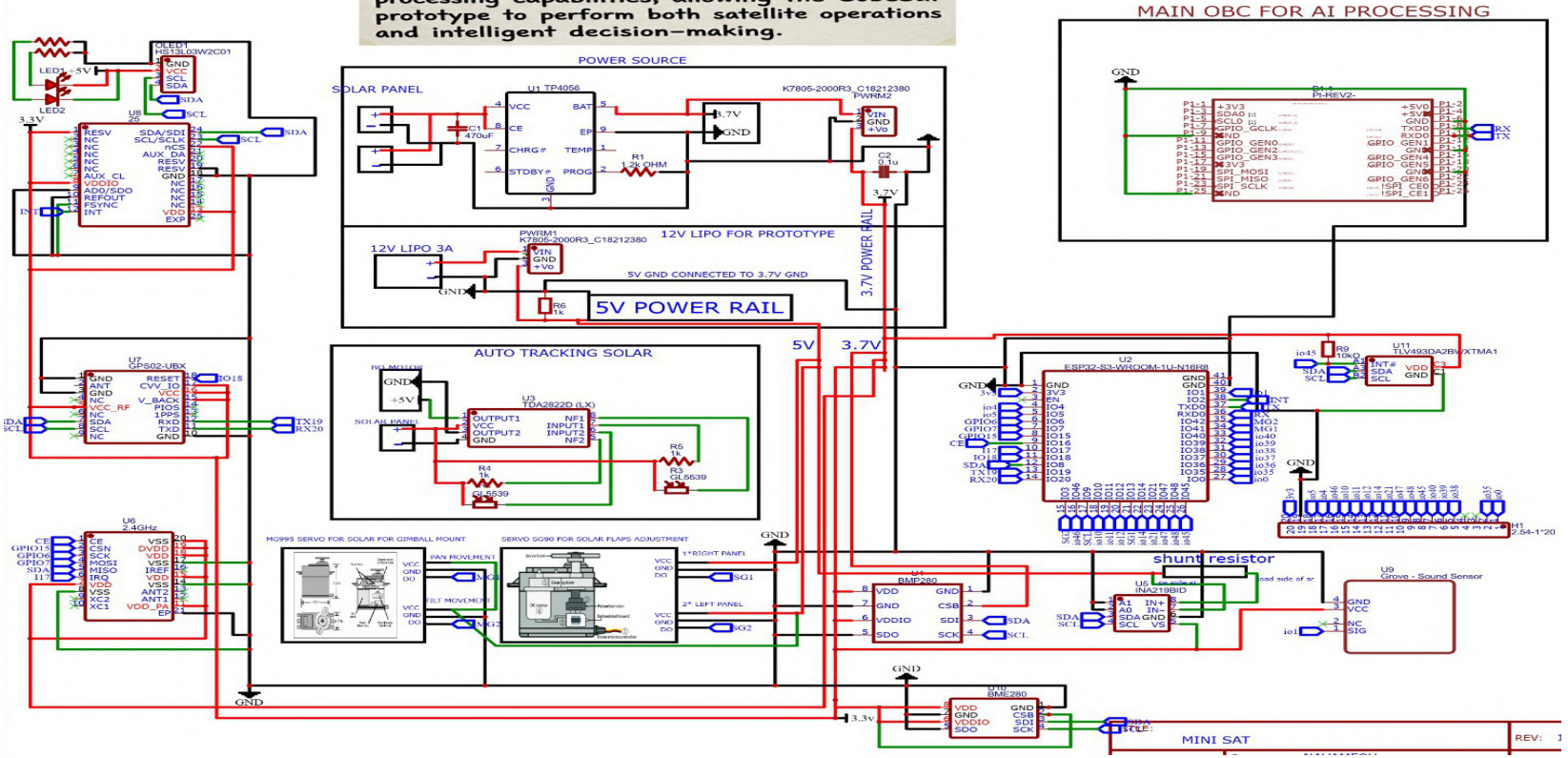
- **Multi-source data fusion –** combines ISRO/MOSDAC satellite data (INSAT LST, SST, rainfall) with IMD gridded rainfall and temperature data into one unified, spatially-aligned dataset.
- **AI-based prediction –**(ConvLSTM/LSTM) Deep Learning based spatio-temporal models generate short-term rainfall and temperature predictions for the pilot region.
- **Interactive geospatial dashboard –** map-based visualization of current and predicted climate states, built on a mapbox.
- **"What-If" scenario simulator –** lets users adjust temperature or rainfall parameters and instantly view simulated downstream impacts on the map.
- **Edge-AI satellite payload (differentiator) –** a custom CubeSat concept with onboard AI (Raspberry Pi) for real-time, compressed telemetry that helps validate the twin.
- **Scalable national framework –** containerized (Docker) architecture designed to extend from one pilot region to pan-India coverage and additional variables.

Process flow diagram or Use-case diagram



Wireframes

The Raspberry OBC– will act as the central onboard computer (OBC) of the AI Space Mission Simulator CubeSat, similar to the computing system used in real spacecraft. It provides a combination of real–time microcontroller control and Linux–based AI processing capabilities, allowing the CubeSat prototype to perform both satellite operations and intelligent decision–making.



Architecture diagram of the proposed solution

POWER SUBSYSTEM

Practical-GaAs Solar Panels
Prototype -18650 Lithium battery
(rechargeable)

POWER MANAGEMENT UNIT

Practical-Solar Array Electronics(SAE)
Prototype-DC-DC buck convertor with 5V 5 amp charging module

RECHARGING UNIT

Prototype- 9*parallel 18650 lion battery ->3s 20A bms-> charging module(input)+output for supply of 12v

3S3P BATTERY PACK
Voltage -11.1v (nominal)
Full charge -12.6v (full charge)

Battery pack connected to bms rated to 3S 20A for stable charging and discharging

Then the main BMS output will be connected to one input(recharge)
+
one output of 12v

Then the output terminals will be connected to DC-DC step down buck converter to power electronics modules

PAYLOAD SUBSYSTEM

MULTI-SPECTRAL CAMERA



Captures images in multiple wavelength and enviornment

- Used for cloud detection
- water analysis
- vegetation monitoring
- land analysis
- quick response imagery

+

MULTIPLE SENSOR FUSION PERF BOARD



To reciprocate as a digital twin satcube

- BMP280 SENSOR
- BME280 SENSOR
- MPU9250 SENSOR
- INA219 SENSOR
- TLV493D SENSOR
- NEO7M GPS SENSOR
- DSB18B20 SENSOR
- SW18020P SENSOR
- TFT 2.4 DISPLAY
- PCA9685 DRIVER

+

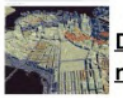
ADCS



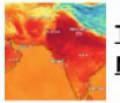
To reciprocate as a onbaord attitude determination & control system

- GYROSCOPE
- ACCELEROMETER
- MAGNETOMETER
- REACTION WHEEL OR SOLENOID SYSTEM

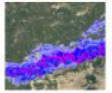
DATA OUTPUT



Digital twin map



Temperature Map



Flood Map



What-if map's

KEY POINTS

- Climate Processing
- What-if scenarios
- AI Processing
- Virtual Digital Map
- Weather Prediction
- Enchances climata data
- Disaster Management
- Quick Disaster Alert
- onboard ai computing reduces data bandwidth

ONBOARD COMPUTER (OBC)

Multiple onboard mcu system

- RASPBERRY PI FOR AI PROCESSING
- STM32-FOR HANDLING OTHER SENSOR PAYLOAD +WORK LOAD





COMMUNICATION SUBSYSTEM(Prototype)



Satcube



MCU



Lora for long distance transmission



COMMAND UPLINK

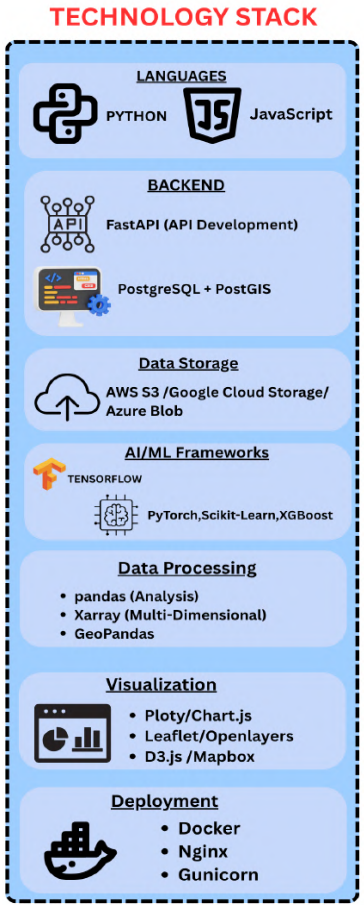
Telemetry Data



- Receives Data
- Sends Command
- Monitor satellite health
- Continuos process Ai edge Ai +Digital twin

Technologies Used in the Solution

- **AI-ML & DL:** SVM, XGBoost, Extra Trees and feature extraction using Extra trees classifier(k=1000) & ConvLSTM / LSTM
- **LIBRARIES USED :** PyTorch & Keras /TensorFlow
- **Data Processing:** Python, pandas, Xarray, GeoPandas, NetCDF / Zarr formats
- **Storage & Database:** PostgreSQL + PostGIS, cloud object storage (Google Cloud Storage / Azure Blob)
- **Backend & Deployment:** FastAPI, Docker, GitHub / CI-CD pipelines
- **Dashboard / Frontend:** React.js, Tailwind CSS, Leaflet / Mapbox for geospatial maps, Plotly / Chart.js for analytics
- **Hardware Payload:** Raspberry Pi (edge AI), STM32 / ESP32-S3 microcontroller, LoRa telemetry module
- **Data Sources:** ISRO MOSDAC (INSAT 3RIMG_L2B_LST / SST / IMC) and IMD gridded rainfall (0.25°×0.25°) and temperature (1.0°×1.0°) datasets



Estimated implementation cost:

Cost Breakdown	
RASPBERRY PI	18699
STM32	2400
LM2596 (BUCK CONVERTER)	199
BMS 3S 20 AMP 18650 Li-ION BATTERY	99
SOLAR PANELS	165
MULTISPECTRAL CAMERA	5672
BMP 280 PRESSURE SENSOR	69
BME280 HUMIDITY SENSOR	269
MPU 9250	549
INA 219 VOLTAGE SENSOR	119
TLV493D MAGNETOMETER	196
NEO 7M GPS SENSOR	716
DSB18B20 TEMPERATURE SENSOR	48
SW18020P VIBRATION SENSOR	40
TFT 2.4 DISPLAY	703
PCA9685 IC	457
LORA MODULE	349
FRAME(3D PRINTING)	2000
MISCELLANEOUS	500
TOTAL	**33249**


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THANK YOU